Big Multiplication

Four digits by two, and an estimate that catches every error.

THE BIG QUESTION

Why does an estimate catch a wrong answer faster than checking the whole calculation again?

HANDS-ON PROJECT

Stadium Count

Estimate a stadium's capacity from a photo, then find the real figure and account for the gap.

FLUENCY GAME

Closest Estimate

Both players estimate before either computes. Closest guess takes the round.

AXIOM · MISSION COMMANDER, NINE TOURS

“Estimate first, every single time. A wrong answer that's roughly right hides. A wrong answer ten times too big cannot.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

NAME ______________

DATE ______________

This is not a test. Nothing here counts for a grade. Two items ask for an estimate rather than an exact answer — give the estimate, not the calculation.

STRAND A
FACTS

A1 7×8

A2 12×6

STRAND B
MULTIPLES OF TEN

B1 40×60

B2 400×70

STRAND C
TWO BY TWO

C1 23×14

C2 45×23

STRAND D
THREE BY TWO

D1 237×45

D2 506×34

STRAND E
ESTIMATING

E1 Estimate 412×19

E2 Estimate 48×52

STRAND F
FINDING ERRORS

F1 Someone says $23 \times 14 = 92$. What did they forget?

F2 Is 34×26 closer to 900 or 9,000?

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Facts	A1 A2	Every Warm-Up tier	___ / 2	Warm-Up drops to 3 items all mission. Facts are assumed from here on.
B · Multiples of ten	B1 B2	Week 2 · lesson 2.1	___ / 2	Day 2.1. This is the Mission 01 skill applied, so a pass here is expected.
C · Two by two	C1 C2	Week 1 · lessons 1.1 to 1.3	___ / 2	Compress Week 1 to two days. Keep 1.4 — the area model is what makes Week 2 land.
D · Three by two	D1 D2	Week 2 · lessons 2.3 and 2.4	___ / 2	Nothing yet. D2 has an interior zero — if that one is right, skip 2.4.
E · Estimating	E1 E2	Week 3 · lessons 3.1 and 3.2	___ / 2	Nothing. Estimating is the Big Question — teach it even if he can already do it.
F · Finding errors	F1 F2	Week 3 · lesson 3.4	___ / 2	Day 3.4. Move the Catch-the-Error set to Challenge only.

ONE NOTE BEFORE YOU START

A fifth grader who can multiply but cannot estimate will not notice a misplaced partial product, and that is the error this mission exists to kill. If strands C and D pass but E fails, do not compress — spend the recovered days on estimation instead.

Two Digits by Two

NAME ___________

DATE ___________

Four partial products, every time. Nothing gets skipped.

Warm-Up

FACTS YOU ALREADY OWN

7 × 8	6 × 9	12 × 5	20 × 30	23 × 10	23 × 4

23 × 14 AS AN AREA · FOUR RECTANGLES, FOUR PARTIAL PRODUCTS

	20	3
10	200	30
4	80	12

Split each number into tens and ones, and the multiplication becomes four rectangles. Add them: 200 + 30 + 80 + 12 = **322**.

The written algorithm does exactly this, but hides the four boxes inside two rows. When somebody gets 92, they found 23 × 4 and forgot the 230 — one whole rectangle missing.

Core

SHOW ALL FOUR PARTIAL PRODUCTS

1. 23 × 14

2. 31 × 22

3. 45 × 23

4. 38 × 24

5. Draw the four-box area model for 38 × 24 and label each box before you add.

Challenge

NOT GRADED

C1. Somebody worked out 45 × 23 and got 135. Which partial products did they find, and which did they miss?

C2. 63 × 48. Estimate first, then compute, then say how close your estimate was.

AXIOM: Count your partial products before you add. Two numbers with two digits each always give four — if you have three, one is missing.

NAME ___________

DATE ___________

Three Digits by Two

Six partial products now. Same rule, one more column.

Warm-Up

200 × 40	30 × 40	7 × 40	200 × 5	30 × 5	7 × 5

Those six are the partial products of 237 × 45. Add them up and you have finished today's first Core item before you started.

237 × 45 · SIX BOXES

	200	30	7
40	8000	1200	280
5	1000	150	35

8000 + 1200 + 280 + 1000 + 150 + 35 = **10,665**.

Estimate first: 240 × 45 is about 10,800. The answer should land just under that, and it does.

Core

ESTIMATE ABOVE EACH ONE FIRST

1. 237 × 45

2. 412 × 19

3. 289 × 31

4. 506 × 34

5. In item 4 the middle digit is a zero. What happens to two of the six boxes, and why?

Challenge

C1. 197 × 203. Estimate it in your head first — the estimate is easier than it looks.

C2. Somebody got 237 × 45 = 1,185. Which partial products are missing from their answer?

NAME ___________

DATE ___________

Carrying Cleanly

Where the carried digit goes, and why it never joins the wrong column.

Warm-Up

SINGLE-DIGIT CARRIES

37 × 4	48 × 6	59 × 7	86 × 5	74 × 8	93 × 9

WHY THE SECOND ROW STARTS ONE PLACE LEFT

```
   45
×  23
-----
  135
  900
-----
 1035
```

The first row is 45 × 3. The second is **not** 45 × 2 — it is 45 × 20, which is why it lands 900 and not 90. That shift left is the whole reason the algorithm works.

A zero in the ones place of the second row makes this visible. Write it in until you no longer need to.

Core

1. 56 × 27

2. 78 × 46

3. 89 × 37

4. 94 × 68

5. In item 2, write the second partial product as a multiplication of 78 by a multiple of ten. Which multiple?

Challenge

C1. Somebody wrote the second row of 56 × 27 as 112 instead of 1120. Their answer is out by how much?

C2. 99 × 99. Do it the long way, then find a shortcut using 100 × 99.

NAME ___________

DATE ___________

The Area Model

Draw it before you write it. The boxes are the proof.

Warm-Up

SPLIT INTO TENS AND ONES

34	67	125	208	450	1,234

DRAW YOUR OWN · 46 × 32

	40	6
30		
2		

Fill the four boxes, then add them below. Check against your estimate: 46 is nearly 50, 32 is about 30, so the answer should be near 1,500.

Core

AREA MODEL, THEN THE ALGORITHM, THEN COMPARE

1. 46 × 32

2. 58 × 43

3. 125 × 24

4. 208 × 36

5. Which method did you find faster, and which one made the answer easier to trust? They need not be the same.

Challenge

C1. Draw the area model for 1,234 × 12. How many boxes does it need?

C2. The area model for 208 × 36 has one box worth nothing. Which one, and what does that tell you about the zero?

Closest Estimate

NAME ___________
DATE ___________

Both players estimate before either one computes.

How to play: one player writes a two-digit by two-digit multiplication. Both players write an estimate — no pencil arithmetic, ten seconds. Then compute the real answer together. Whoever's estimate is closer wins the round. An estimate that took longer than ten seconds does not count.

Warm-Up · round to the nearest ten first

Estimate 19 × 21, then compute it

Estimate 48 × 52, then compute it

Play six rounds

ROUND	PROBLEM	MY ESTIMATE	REAL ANSWER	OFF BY
1				
2				
3				
4				
5				
6				

Challenge

C1. Rounding both numbers up always overestimates. Rounding one up and one down does what? Try 48 × 52 and see.

C2. Which of your six rounds had the worst estimate? What made that one harder to guess?

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ · end of week 2
TEST · end of week 4

Week 2 · Bigger Numbers QUIZ FRIDAY

Four digits by two, and the estimate stops being optional. Every Core item gets an estimate written above it.

MON 2.1
Multiples of ten first
40 × 60 before 43 × 67. Every time.

TUE 2.2
Estimate, then compute
Round both, multiply, then do it properly and compare.

WED 2.3
Four digits by two
Longer, not harder. Keep the estimate in your head.

THU 2.4
Catch the error
Given a wrong worked answer, find the line it went wrong.

FRI
Mid-unit quiz
Page 10. Eight items, 85% to keep flying.

Week 3 · Estimation as a Habit

The mission's real content. An estimate is not a worse answer — it is the thing that tells you whether the real answer is believable.

MON 3.1
Round to estimate
Which way to round, and what it costs you.

TUE 3.2
Over or under
Say before you compute whether your estimate is high or low.

WED 3.3
Order of magnitude
Is it hundreds, thousands or tens of thousands?

THU 3.4
Mark somebody's work
Six worked answers, three wrong. Find them by estimating.

FRI
Stadium Count begins
Estimate from the photo before looking anything up.

Week 4 · Proof UNIT TEST

The Stadium Count gets finished and the gap between estimate and reality gets explained out loud.

MON 4.1
Count a section
Seats per row, rows per section, sections per stadium.

TUE 4.2
Scale it up
One section times the number of sections.

WED 4.3
Explain the gap
Compare to the published capacity and account for the difference.

THU
Error journal sweep
Fix only what repeats.

FRI
Mission 02 test
Pages 11–12. Twelve items plus one explanation.

NAME ______________

SCORE ______ / 8

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. 40 × 60

2. 23 × 14

3. 45 × 23

4. 237 × 45

5. Estimate 412 × 19

6. 1,234 × 12

7. 506 × 34

8. Somebody says 34 × 26 = 68. Without computing the real answer, explain how an estimate tells you immediately that this is wrong.

Marking note: item 8 wants the reasoning, not the product. Accept anything along the lines of “30 × 30 is about 900, so an answer of 68 is far too small.”

Unit Test · Part 1

NAME ______________

DATE ______________

Items 1–8. No time limit. Write an estimate above every calculation.

1. 50×50 ....
2. 38×24 ....
3. 56×27 ....
4. 289×31 ....
5. 125×24 ....
6. $2{,}450 \times 36$ ....
7. Estimate $5{,}120 \times 48$ ....
8. Estimate 197×203 ....

CONTINUES ON PAGE 12

Unit Test · Part 2

Items 9–12 and the Big Question.

SCORE ______ / 12

BAND ______________

9. 1,875 × 24

10. 9,999 × 99

11. Somebody worked out 237 × 45 and got 1,185. Which partial products did they miss, and what is the right answer?

12. A stadium has 38 rows of 24 seats in each of 16 sections. Estimate the capacity first, then work it out exactly.

EXPLAIN YOUR THINKING · THE BIG QUESTION

Why does an estimate catch a wrong answer faster than checking the whole calculation again?

Use one wrong answer from this mission as your example. Scored on reasoning, not neatness.

11–12
Gold band. The Scale Badge is earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 56 | A2 · 72 | B1 · 2400 | B2 · 28,000

C1 · 322 | C2 · 1035 | D1 · 10,665 | D2 · 17,204

E1 · about 8,000 | E2 · about 2,500 | F1 · the 230 | F2 · 900

Lesson 1.1 · Two Digits by Two

Warm-Up · 56 | 54 | 60 | 600 | 230 | 92

Core · 1 · 322 | 2 · 682 | 3 · 1035 | 4 · 912 | 5 · 600 + 120 + 160 + 32 = 912

C1 · they found 45 × 3 = 135 and stopped; the missing 900 is 45 × 20. C2 · 3,024, with an estimate of 60 × 50 = 3,000.

Lesson 1.2 · Three Digits by Two

Warm-Up · 8000 | 1200 | 280 | 1000 | 150 | 35

Core · 1 · 10,665 | 2 · 7,828 | 3 · 8,959 | 4 · 17,204

Item 5 · both boxes in the tens column are worth zero, so four of the six boxes carry the whole answer. C1 · about 40,000 — 200 × 200. C2 · they found 237 × 5 only; the three missing boxes are 237 × 40, worth 9,480.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · Carrying Cleanly

Warm-Up · 148 | 288 | 413 | 430 | 592 | 837

Core · 1 · 1512 | 2 · 3588 | 3 · 3293 | 4 · 6392 | 5 · 78 × 40 = 3120

C1 · out by 1,008 — they wrote 56 × 2 instead of 56 × 20. C2 · 9,801; the shortcut is 100 × 99 = 9,900 then subtract one 99.

Lesson 1.4 · The Area Model

Warm-Up · 30+4 | 60+7 | 100+20+5 | 200+8 | 400+50 | 1000+200+30+4

Core · 1 · 1472 | 2 · 2494 | 3 · 3000 | 4 · 7488

Item 5 · accept either answer with a reason; most students find the algorithm faster and the area model more trustworthy, which is the point. C1 · eight boxes, since 1,234 splits four ways and 12 splits two. C2 · the tens box is zero, because 208 has no tens — the zero holds the place open without adding anything.

Friday · Closest Estimate

Warm-Up · 19 × 21 ≈ 400, actually 399 | 48 × 52 ≈ 2500, actually 2496

C1 · rounding one up and one down largely cancels out, which is why 48 × 52 estimates so well — worth pointing out that this is not luck. C2 · his own answer; look for an explanation mentioning digits near five, which round furthest.

TEACHER PAGE · ANSWER KEY

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 2400
2 · 322
3 · 1035
4 · 10,665
5 · about 8,000
6 · 14,808
7 · 17,204
8 · reasoning

Unit test

1 · 2500
2 · 912
3 · 1512
4 · 8,959
5 · 3000
6 · 88,200
7 · about 250,000
8 · about 40,000
9 · 45,000
10 · 989,901

Items 11, 12 and the Big Question

11 · they found only 237 × 5; the missing 9,480 is 237 × 40, and the right answer is 10,665. **12** · estimate 40 × 25 × 16 = 16,000; exactly 38 × 24 × 16 = 14,592. **Big Question** · full marks for any answer making the point that an estimate checks the size of the answer rather than the steps, so a missing partial product or a misplaced digit shows up as an answer ten times too small — whereas re-checking the calculation tends to repeat the same mistake.

Error journal · Mission 02

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY